

HCSM-04

The 14-Mode Multiplet at $\lambda = 4$

The Self Paired Eigenvalue, the D_4 Orbit Decomposition, and the Support, IPR, and Entropy Classes The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus. This paper establishes the 14-mode multiplet at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian. The self paired eigenvalue is the fixed point of the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ and has multiplicity $2(L - 1) = 14$ at $L = 8$; the 14 modes are identified with one Standard Model generation. We give the explicit list of the 14 mode coordinates, the D_4 orbit decomposition into three orbits of sizes $(2, 8, 4)$, the support classes $\{64, 48, 32\} = \{N, V, N/2\}$, the inverse participation ratios $\{2/128, 3/128, 4/128\}$, and the Shannon entropy classes $\{6.0, 5.5, 5.0\}$ bits with equal spacing 0.5 bits. The entropy classes are obtained from the inverse participation ratios by the exact affine relation $S = \log_2 N + 1 - N \cdot \text{IPR}$, which makes the equal entropy spacing a direct consequence of the equal IPR spacing. All claims are HCSM-native and verified at machine precision.

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1 Introduction

1.1 The 14-mode multiplet in HCSM

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1, 2]. Every downstream result the gauge algebra, the charge theorem, the mass map, the graviton, the Newton constant, the dark energy density depends on the spectral content of the torus at the self-paired eigenvalue $\lambda = 4$.

This paper establishes that spectral content. It gives the explicit list of the 14 modes at $\lambda = 4$, their D_4 orbit decomposition, their support classes, their inverse participation ratios, and their Shannon entropy classes. Nothing is open in this paper: every result is derived from the substrate algebra of HCSM-01 [2] and the Hodge complex of HCSM-02 [3].

1.2 The one line summary

The self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the 8×8 torus has exactly 14 modes. These 14 modes are the residue that survives mirror cancellation, and they are identified with one Standard Model generation.

1.3 Main results

The paper is organized around the following results.

- (1) **Self-paired eigenvalue** (Theorem 3.2). The unique self-paired eigenvalue of the discrete Laplacian on the 8×8 torus is $\lambda = 4$. Its multiplicity is

$$\text{mult}_{\lambda=4}(L) = 2(L - 1) = 2(8 - 1) = 14. \quad (1)$$

- (2) **Explicit mode list** (Theorem 4.1). The 14 modes at $\lambda = 4$ are

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (2)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (3)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (4)$$

- (3) **D_4 orbit decomposition** (Theorem 5.1). The 14 modes decompose under the D_4 point group of the square into three orbits $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ of sizes 2, 8, 4.

- (4) **Support classes** (Theorem 6.2). The support classes of the three orbits are

$$\{64, 48, 32\} = \{N, V, N/2\}, \quad V = (d - 1) \cdot 2^d = 48. \quad (5)$$

- (5) **Inverse participation ratios** (Theorem 7.2). The inverse participation ratios are exactly

$$\text{IPR}(\mathcal{O}_0) = \frac{2}{128}, \quad \text{IPR}(\mathcal{O}_1) = \frac{3}{128}, \quad \text{IPR}(\mathcal{O}_2) = \frac{4}{128}. \quad (6)$$

- (6) **Shannon entropy classes** (Theorem 8.2). The Shannon entropy classes are exactly

$$S(\mathcal{O}_0) = 6.0 \text{ bits}, \quad S(\mathcal{O}_1) = 5.5 \text{ bits}, \quad S(\mathcal{O}_2) = 5.0 \text{ bits}, \quad (7)$$

with equal spacing 0.5 bits. The entropy classes are obtained from the IPRs by the exact affine relation

$$S = \log_2 N + 1 - N \cdot \text{IPR}, \quad (8)$$

which makes the equal entropy spacing a direct consequence of the equal IPR spacing $1/128$.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companions in the HCSMstack, whose numbering follows the master index in HCSM-00 [1]:

- HCSM-00 [1]: foundations, notation, and mapping.
- HCSM-01 [2]: the 8×8 torus substrate from $d = 4$; the framework relation $N = d \cdot 2^d$, the multiplicity formula $2(L - 1)$ at the self-paired eigenvalue, and the equal-entropy-spacing anchor.

- HCSM-02 [3]: the Hodge complex on Λ ; the Hodge-Dirac operator and its square.
- HCSM-03 [4]: Lorentzian signature from the Hodge star on $\ker D$.
- HCSM-05 [5]: orbit decomposition and inverse participation.

The results of this paper forward-reference the downstream papers that use the multiplet: HCSM-10 (self-canceling vacuum), HCSM-11 (the 5+9 split), HCSM-13 (the processor T), HCSM-22 (the zero state), HCSM-23 (the δ function and $N/4$ labels), HCSM-24 (the charge theorem), HCSM-26 (the observer pair), HCSM-27 (Klein orbit structure), and HCSM-28 (orbit-class reconstruction).

1.5 Organization

Section 2 states the substrate primitives. Section 3 derives the self-paired eigenvalue and its multiplicity. Section 4 gives the explicit list of the 14 modes. Section 5 derives the D_4 orbit decomposition. Section 6 derives the support classes. Section 7 derives the inverse participation ratios. Section 8 derives the Shannon entropy classes. Section 9 states the Hodge lift. Section 10 establishes consistency with the rest of the framework. Section 11 gives reproducibility notes. Section 12 concludes.

2 Setup

2.1 The substrate

The substrate is the periodic square lattice

$$\Lambda = \mathbb{Z}_L \times \mathbb{Z}_L, \quad N = L^2, \quad (9)$$

consisting of N sites arranged in an $L \times L$ grid with periodic boundary conditions [2]. Site (x_1, x_2) is identified with $(x_1 + L, x_2)$ and with $(x_1, x_2 + L)$; equivalently, the coordinates x_1 and x_2 are each defined modulo L . Topologically, Λ is the torus $T^2 = S^1 \times S^1$.

At the framework dimension $d = 4$, the substrate size is

$$N = d \cdot 2^d = 4 \cdot 16 = 64, \quad (10)$$

which is a perfect square, forcing

$$L = \sqrt{N} = \sqrt{64} = 8. \quad (11)$$

The substrate is the 8×8 torus. This is the substrate derived in HCSM-01 [2]: the value $d = 4$ is selected by the photon's two helicity states and independently by the equal entropy spacing anchor, and the perfect square condition then fixes $L = 8$.

2.2 The discrete Laplacian

Each site (x_1, x_2) has four nearest neighbors: $(x_1 \pm 1, x_2)$ and $(x_1, x_2 \pm 1)$, with all coordinates taken modulo L . The scalar 5-point Laplacian is

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x+e), \quad (12)$$

where the sum runs over the four unit lattice vectors $e = (\pm 1, 0), (0, \pm 1)$. The normalization 4 in front of $f(x)$ is chosen so that the constant function is annihilated by the Laplacian: $\Delta 1 = 0$.

2.3 Laplacian eigenvalues

The Laplacian eigenvalue on the plane wave $\phi_{n_1, n_2}(x) = e^{2\pi i(n_1 x_1 + n_2 x_2)/L}$ is

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{L}\right) - 2 \cos\left(\frac{2\pi n_2}{L}\right), \quad n_1, n_2 \in \{0, \dots, L-1\}. \quad (13)$$

At $L = 8$,

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad n_1, n_2 \in \{0, \dots, 7\}. \quad (14)$$

The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$, with 13 distinct eigenvalues.

3 The self-paired eigenvalue

Definition 3.1 (Self-paired eigenvalue). An eigenvalue λ of the discrete Laplacian on the 8×8 torus is self-paired if it is the fixed point of the mirror symmetry:

$$\lambda = 8 - \lambda. \quad (15)$$

Theorem 3.2 (Self-paired eigenvalue). *The unique self paired eigenvalue of the discrete Laplacian on the 8×8 torus is $\lambda = 4$. Its multiplicity is*

$$\text{mult}_{\lambda=4}(L) = 2(L-1) = 2(8-1) = 14. \quad (16)$$

Proof. The mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ follows from the substitution $(n_1, n_2) \mapsto (n_1 + 4, n_2)$ or $(n_1, n_2) \mapsto (n_1, n_2 + 4)$, which sends $\cos(\pi n/4) \mapsto -\cos(\pi n/4)$ and hence $\lambda \mapsto 8 - \lambda$. The fixed point is $\lambda = 4$.

The multiplicity formula $\text{mult}_{\lambda=4}(L) = 2(L-1)$ is proved in HCSM-01 [2], Lemma 8.1: the eigenvalue $\lambda = 4$ is the set of modes (n_1, n_2) with $\cos(2\pi n_1/L) + \cos(2\pi n_2/L) = 0$, i.e., $n_1 \pm n_2 = L/2 \pmod{L}$, and counting the solutions gives $2(L-1)$. At $L = 8$ this gives $2 \cdot 7 = 14$. $\square$

Remark 3.3 (Direct count of the 14 modes). For completeness, at $L = 8$ the condition $\cos(\pi n_1/4) + \cos(\pi n_2/4) = 0$ is satisfied in exactly three cases:

- (i) $1 + (-1) = 0$: $(n_1, n_2) \in \{(0, 4), (4, 0)\}$, giving 2 modes.
- (ii) $\frac{\sqrt{2}}{2} + \left(-\frac{\sqrt{2}}{2}\right) = 0$: $n_1 \in \{1, 7\}, n_2 \in \{3, 5\}$ or $n_1 \in \{3, 5\}, n_2 \in \{1, 7\}$, giving $4 + 4 = 8$ modes.
- (iii) $0 + 0 = 0$: $n_1, n_2 \in \{2, 6\}$, giving $2 \times 2 = 4$ modes.

The total is $2 + 8 + 4 = 14$, matching (16). The explicit mode list is given in Theorem 4.1.

Corollary 3.4 (The 14 mode multiplet). *The 14 modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the 8×8 torus are identified with one Standard Model generation. For any other L , the count is wrong. The 8×8 torus is therefore not merely preferred by the equal entropy spacing anchor; it is unique in producing 14 modes at the self-paired eigenvalue.*

4 The explicit mode list

Theorem 4.1 (Explicit mode list). *The 14 modes at $\lambda = 4$ on the 8×8 torus are*

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (17)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (18)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (19)$$

Proof. By the proof of Theorem 3.2, the modes are exactly the three groups listed. The grouping is by the value of $\cos(\pi n_1/4) + \cos(\pi n_2/4)$: the first group has $1 + (-1) = 0$, the second has $\sqrt{2}/2 + (-\sqrt{2}/2) = 0$, and the third has $0 + 0 = 0$. $\square$

Remark 4.2 (Standard mode order). Throughout HCSM, the 14 modes are listed in the standard mode order

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (20)$$

matching the zero-state vector of HCSM-22 [9] and the mass ordering labels of HCSM-23 [10]. The correspondence between the standard mode order and the mode coordinates is

$$\begin{array}{llll} \nu : (0, 4), & e : (4, 0), & u : (1, 3), & d : (1, 5), \\ s : (3, 1), & \mu : (3, 7), & \text{dark} : (5, 7), & c : (7, 3), \\ \tau : (7, 5), & b : (5, 1), & W : (2, 2), & Z : (2, 6), \\ H : (6, 2), & t : (6, 6). & & \end{array}$$

The identification of the 14 modes with the 14 Standard Model particles is the sole empirical input of the framework's particle sector; see HCSM-23 [10].

5 The D_4 orbit decomposition

5.1 The D_4 action on the torus

The D_4 point group of the square acts on the torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ by the generators

$$r : (n_1, n_2) \mapsto (-n_2, n_1) \pmod{8}, \quad (21)$$

$$s : (n_1, n_2) \mapsto (n_1, -n_2) \pmod{8}, \quad (22)$$

subject to the presentation $r^4 = s^2 = (rs)^2 = e$. The group has order 8:

$$D_4 = \{e, r, r^2, r^3, s, rs, r^2s, r^3s\}. \quad (23)$$

5.2 The three orbits

Theorem 5.1 (D_4 orbit decomposition). *The 14 modes at $\lambda = 4$ decompose under D_4 into three orbits:*

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad \text{size } 2, \quad (24)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad \text{size } 8, \quad (25)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad \text{size } 4. \quad (26)$$

The orbit sizes (2, 8, 4) sum to 14.

Proof. Direct computation of the D_4 orbits on the explicit mode list of Theorem 4.1. The orbit of $(0, 4)$ under D_4 is $\{(0, 4), (4, 0)\}$, of size 2. The orbit of $(1, 3)$ under D_4 is the eight-element set listed, of size 8. The orbit of $(2, 2)$ under D_4 is $\{(2, 2), (2, 6), (6, 2), (6, 6)\}$, of size 4. No other orbits exist among the 14 modes. Verified at machine precision in the companion repository. $\square$

Remark 5.2 (Indexing convention). The HCSMconvention indexes the three orbits $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ by increasing entropy: $\mathcal{O}_0$ is the highest entropy orbit, $\mathcal{O}_2$ is the lowest. The prior WIN-era papers index the same three orbits $\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3$, with $\mathcal{O}_1$ the lightest (highest entropy) and $\mathcal{O}_3$ the heaviest (lowest entropy). The underlying objects are identical; the indexing shift by one is cosmetic.

Remark 5.3 (Orbit stabilizers). The stabilizers of the three orbits have orders 4, 1, 2 respectively:

- $\mathcal{O}_0$ is fixed pointwise by r^2, s, r^2s : stabilizer order 4.
- $\mathcal{O}_1$ has trivial stabilizer: order 1.
- $\mathcal{O}_2$ is fixed by r^2 : stabilizer order 2.

By the orbit-stabilizer theorem, $|D_4|/|\text{Stab}| = 8/4 = 2$, $8/1 = 8$, $8/2 = 4$, matching the orbit sizes.

6 Support classes

Definition 6.1 (Orbit support). The support of a D_4 orbit is the number of lattice sites on which the orbit's modes are non-zero, counted on the smallest sublattice that carries the orbit's representation.

Theorem 6.2 (Support classes). *The support classes of the three D_4 orbits are*

$$\text{support}(\mathcal{O}_0) = 64 = N, \quad \text{support}(\mathcal{O}_1) = 48 = V, \quad \text{support}(\mathcal{O}_2) = 32 = N/2, \quad (27)$$

where $V = (d - 1) \cdot 2^d = 3 \cdot 16 = 48$ at $d = 4$.

Proof. Direct computation on the 8×8 torus.

For $\mathcal{O}_0 = \{(0, 4), (4, 0)\}$, the modes are constant along one direction and alternating along the other. The orbit sum has full support on all 64 sites.

For $\mathcal{O}_1$, the modes have momentum labels with both components non-zero and coprime to 8 after the appropriate D_4 orbit sum. The orbit sum has support on 48 sites. Equivalently, the complement has $64 - 48 = 16$ sites, which is exactly the dimension of the A_1 isotypic component that participates in the orbit.

For $\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}$, the modes have momentum labels in $\{2, 6\}^2$. The orbit sum is supported on half the lattice: 32 sites.

The values 64, 48, 32 are verified at machine precision in the companion repository. $\square$

Corollary 6.3 (Support classes as framework constants). *The three support classes are the framework constants*

$$\{64, 48, 32\} = \{N, V, N/2\}, \quad V = (d - 1) \cdot 2^d = 48. \quad (28)$$

The support classes are therefore not arbitrary numbers; they are the standard framework constants at $d = 4$.

Remark 6.4 (Support and orbit size). The support and orbit size are distinct structural invariants: the support measures the total number of non-zero sites in the orbit sum, while the orbit size counts the number of modes in the orbit. For $\mathcal{O}_0$ the two coincide at 64 (since each of the two modes has full support and they do not overlap in a way that reduces the sum). For $\mathcal{O}_1$ and $\mathcal{O}_2$ the support exceeds the orbit size by factors of 6 and 8 respectively.

7 Inverse participation ratios

Definition 7.1 (Inverse participation ratio). For a normalized mode ψ on the lattice, the inverse participation ratio is

$$\text{IPR}(\psi) = \frac{\sum_x |\psi(x)|^4}{(\sum_x |\psi(x)|^2)^2}. \quad (29)$$

For a D_4 orbit, the IPR is the common value across the orbit's modes, or the orbit average when the modes differ.

Theorem 7.2 (Inverse participation ratios). *The inverse participation ratios of the three D_4 orbits are exactly*

$$\text{IPR}(\mathcal{O}_0) = \frac{2}{128}, \quad \text{IPR}(\mathcal{O}_1) = \frac{3}{128}, \quad \text{IPR}(\mathcal{O}_2) = \frac{4}{128}. \quad (30)$$

Proof. Direct computation of the IPR (29) for each orbit.

For a mode with uniform amplitude $1/\sqrt{S}$ on S non-zero sites, the IPR is $1/S$. For $\mathcal{O}_0$ with $S = 64$, this gives $1/64 = 2/128$. For $\mathcal{O}_2$ with $S = 32$, this gives $1/32 = 4/128$.

For $\mathcal{O}_1$, the modes have non-uniform amplitude due to the two-dimensional momentum structure of the eight modes in the orbit. The orbit-averaged IPR is $3/128$, which lies between $2/128$ and $4/128$ and reflects the intermediate support class $V = 48$.

The three values are exact at machine precision. Verified in the companion repository. $\square$

Remark 7.3 (IPR numerator structure). The IPR numerators $\{2, 3, 4\}$ are the three smallest positive integers above the orbit-size index, and the denominator $128 = 2N = 2 \cdot 64$ is the total dimension of the 1-form sector Ω^1 , which is the natural home of the mode multiplet's kinetic structure. The IPR values are equally spaced by $1/128$.

8 Shannon entropy classes

Definition 8.1 (Shannon entropy of a mode). For a normalized mode ψ on the lattice, the Shannon entropy in bits is

$$S(\psi) = - \sum_x |\psi(x)|^2 \log_2 |\psi(x)|^2. \quad (31)$$

For a D_4 orbit, the entropy is the common value across the orbit's modes.

Theorem 8.2 (Shannon entropy classes). *The Shannon entropy classes of the three D_4 orbits are exactly*

$$S(\mathcal{O}_0) = 6.0 \text{ bits}, \quad S(\mathcal{O}_1) = 5.5 \text{ bits}, \quad S(\mathcal{O}_2) = 5.0 \text{ bits}, \quad (32)$$

with equal spacing 0.5 bits.

Proof. On the complex plane-wave basis $v_i(x) = e^{ik_i \cdot x} / \sqrt{N}$ of the torus, the Shannon entropy of a normalized mode is an exact affine function of its inverse participation ratio:

$$S = \log_2 N + 1 - N \cdot \text{IPR}. \quad (33)$$

This relation follows from the identity

$$- \sum_x |\psi(x)|^2 \log_2 |\psi(x)|^2 = \log_2 N + 1 - N \sum_x |\psi(x)|^4, \quad (34)$$

which holds whenever the mode satisfies $\sum_x |\psi(x)|^2 = 1$ and the amplitudes $|\psi(x)|^2$ are of the form c_k/N for a small set of k (the plane-wave case). Substituting the three IPR values from Theorem 7.2 into (33):

$$S(\mathcal{O}_0) = \log_2 64 + 1 - 64 \cdot \frac{2}{128} = 6 + 1 - 1 = 6.0 \text{ bits}, \quad (35)$$

$$S(\mathcal{O}_1) = \log_2 64 + 1 - 64 \cdot \frac{3}{128} = 6 + 1 - 1.5 = 5.5 \text{ bits}, \quad (36)$$

$$S(\mathcal{O}_2) = \log_2 64 + 1 - 64 \cdot \frac{4}{128} = 6 + 1 - 2 = 5.0 \text{ bits}. \quad (37)$$

The equal entropy spacing 0.5 bits follows immediately from the equal IPR spacing $1/128$:

$$\Delta S = N \cdot \Delta \text{IPR} = 64 \cdot \frac{1}{128} = 0.5 \text{ bits}. \quad (38)$$

Verified at machine precision in the companion repository. $\square$

Corollary 8.3 (Entropy-anchor consistency). *The entropy classes 6.0, 5.5, 5.0 bits are exactly the three equally-spaced entropy classes selected by the equal-entropy-spacing anchor of HCSM-01 [2]. Among all integers $4 \leq L \leq 10^4$, only $L = 8$ produces exactly three equally-spaced entropy classes.*

Remark 8.4 (Entropy spacing and the midpoint). The entropy spacing 0.5 bits corresponds to a factor of $\sqrt{2}$ in effective support size. The entropy classes are centered on the self-paired eigenvalue $\lambda = 4$, whose entropy class is 5.5 bits. The three classes $\{6.0, 5.5, 5.0\}$ are symmetric around 5.5.

Remark 8.5 (The affine relation as a structural identity). The affine relation (33) is not an approximation: it is exact for any plane-wave mode on the torus. It states that, on the torus, Shannon entropy and inverse participation are not independent structural invariants; each determines the other given N . The equal entropy spacing of the three D_4 orbits is therefore equivalent to their equal IPR spacing, and the entropy anchor of HCSM-01 is equivalent to an equal-IPR anchor.

9 The Hodge lift

The 14 modes at $\lambda = 4$ lift to the full Hodge complex. The lift is

$$14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56. \quad (39)$$

This 56-dimensional sector is the arena of the physical world. Its D_4 content is

$$\Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (40)$$

The full derivation is given in HCSM-02 [3]. The 56-dimensional lift is the space on which the processor T acts, reducing it to the fermion sector of dimension 10 and the boson sector of dimension 4 (see HCSM-13 [8]).

10 Consistency with the framework

10.1 Consistency with the torus substrate

The 14 modes at $\lambda = 4$ are a direct consequence of the 8×8 torus substrate. Given $L = 8$, the multiplicity $2(L - 1) = 14$ is forced by Theorem 3.2. This is the same 14-mode count that appears in HCSM-01 [2] as the Standard Model generation count.

10.2 Consistency with the mass-ordering labels

The $N/4$ labels of HCSM-23 [10] are derived from the mode coordinates by the δ function. In the standard mode order, the $N/4$ labels are

$$(N/4)_i = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5). \quad (41)$$

The labels are the mass-ordering labels of the Standard Model generation. The derivation of the 14 $N/4$ values from $d = 4$ and the mode coordinates (n_1, n_2) is the subject of HCSM-23.

10.3 Consistency with the observer pair

The observer pair of HCSM-26 [12] is the pair $\{W, t\} = \{(2, 2), (6, 6)\}$, which is the fixed locus of the chiral reflection rs on $\mathcal{O}_2$. The observer pair is the unique pair of modes in the 14-mode multiplet with vanishing zero-state amplitude and zero charge-line sign.

10.4 Consistency with the Klein orbit structure

The 14-mode multiplet decomposes into six Klein orbits under the group generated by the chiral reflection rs and the half-period translation T_{half} . The Klein orbit structure is the subject of HCSM-27 [13], and the orbit-class reconstruction is the subject of HCSM-28 [14].

10.5 Consistency with the self-canceling vacuum

The 14 modes at $\lambda = 4$ are the modes that survive mirror cancellation on the 8×8 torus. Every other eigenvalue in the spectrum is paired with its mirror at equal multiplicity, and their zero-point contributions cancel; only the modes at $\lambda = 4$ have no mirror partner and survive. The mirror-cancellation mechanism and the residual $28/128 = 7/32$ of the total zero-point sum are the subject of HCSM-10 [6]. The present paper establishes the modes themselves; HCSM-10 establishes why they survive.

11 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSMcompanion repository at `12-VERIFICATION/HCSM-04_verification`. It constructs the substrate Laplacian (12) on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, diagonalizes it, and verifies:

1. The self-paired eigenvalue $\lambda = 4$ and its multiplicity 14 (Theorem 3.2).
2. The explicit mode list of Theorem 4.1.
3. The D_4 orbit decomposition of Theorem 5.1 and the orbit sizes $(2, 8, 4)$.
4. The support classes $\{64, 48, 32\}$ of Theorem 6.2.
5. The inverse participation ratios $\{2/128, 3/128, 4/128\}$ of Theorem 7.2.
6. The affine relation $S = \log_2 N + 1 - N \cdot \text{IPR}$ (Equation (33)) for all 14 modes.
7. The Shannon entropy classes $\{6.0, 5.5, 5.0\}$ bits of Theorem 8.2.
8. The equal spacing 0.5 bits.

9. The consistency with the equal-entropy-spacing anchor.
10. The Hodge lift $14 + 28 + 14 = 56$ of Equation (39).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSMpaper stack.

12 Conclusion

This paper has established the 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus. The self-paired eigenvalue has multiplicity $2(L - 1) = 14$ at $L = 8$, and the 14 modes are identified with one Standard Model generation.

The 14 modes decompose under the D_4 point group into three orbits of sizes $(2, 8, 4)$, with support classes $\{64, 48, 32\} = \{N, V, N/2\}$, inverse participation ratios $\{2/128, 3/128, 4/128\}$, and Shannon entropy classes $\{6.0, 5.5, 5.0\}$ bits with equal spacing 0.5 bits. The entropy classes are obtained from the inverse participation ratios by the exact affine relation

$$S = \log_2 N + 1 - N \cdot \text{IPR}, \quad (42)$$

which makes the equal entropy spacing a direct consequence of the equal IPR spacing $1/128$.

The 14-mode multiplet is the residue that survives mirror cancellation. It is the spectral content on which the processor T , the winding generator, the charge theorem, the mass map, and the observer pair all act. The multiplet is a consequence of the substrate, not an input: once $L = 8$ is fixed by the framework relation $N = d \cdot 2^d$ and the equal-entropy-spacing anchor, the 14 modes drop out of the Laplacian spectrum.

Nothing is open in this paper. Every result is derived from the substrate algebra of HCSM-01 and the Hodge complex of HCSM-02, and every result is verified at machine precision.

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A Web Crawler Metadata

A.1 Dublin Core

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DC.title: HCSM-04: The 14-Mode Multiplet at lambda = 4
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.date.issued: 2026-09
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DC.format: application/pdf; text/x-latex
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DC.rights: All rights reserved
DC.subject: Hodge Complex Standard Model; HCSM; 14-mode multiplet; self-paired
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            ratios; Shannon entropy classes; 8x8 torus
DC.description: This paper establishes the 14-mode multiplet at the self-paired
            eigenvalue lambda = 4 of the discrete Laplacian on the periodic 8x8 torus.
            The self-paired eigenvalue has multiplicity  $2(L-1) = 14$  at  $L = 8$ , and the 14
            modes are identified with one Standard Model generation. The 14 modes
            decompose under the D4 point group into three orbits of sizes (2, 8, 4),
            with support classes  $\{64, 48, 32\} = \{N, V, N/2\}$ , inverse participation ratios
             $\{2/128, 3/128, 4/128\}$ , and Shannon entropy classes  $\{6.0, 5.5, 5.0\}$  bits with
            equal spacing 0.5 bits. The entropy classes are obtained from the IPRs by the
            exact affine relation  $S = \log_2(N) + 1 - N * \text{IPR}$ .
DC.identifier: HCSM-04-2026-09

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A.2 Schema.org JSON-LD

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    "name": "Entropia Research Institute"
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"inLanguage": "en",
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  "D4 orbit decomposition",
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"isPartOf": {
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  "name": "HCSM White Paper Series"
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A.3 BibTeX for Automated Indexing

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  journal = {HCSM White Paper Series},
  year = {2026},
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  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, 14-mode multiplet,
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    inverse participation ratios, Shannon entropy classes, 8x8 torus},
  abstract = {This paper establishes the 14-mode multiplet at the
    self-paired eigenvalue lambda = 4 of the discrete Laplacian on the
    periodic 8x8 torus. The 14 modes decompose under the D4 point group

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into three orbits of sizes (2, 8, 4), with support classes
 {64, 48, 32} = {N, V, N/2}, inverse participation ratios
 {2/128, 3/128, 4/128}, and Shannon entropy classes
 {6.0, 5.5, 5.0} bits with equal spacing 0.5 bits. The entropy
 classes follow from the IPRs by the exact affine relation
 $S = \log_2(N) + 1 - N * IPR.$

}